Electric Vehicle Sales Analysis in India

Internship Project Report
Unified Mentor Internship Program

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1. Introduction

The automotive industry in India is undergoing a major transformation with the increasing adoption of Electric Vehicles (EVs). Driven by government incentives, rising fuel costs, and environmental awareness, the EV market has witnessed significant growth in recent years.

This project analyses EV sales data from 2014 to 2024 across various states of India. The analysis explores sales trends over time, regional performance, and the distribution of sales among different vehicle categories. The insights derived can help policymakers, manufacturers, and investors make informed decisions in the EV sector.

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2. Objectives

The main objectives of this analysis are:

1. To track EV sales growth over the last decade.
2. To identify the top-performing states in EV adoption.
3. To analyse EV sales trends by vehicle type and category.
4. To understand the overall market distribution and consumer preferences.
5. To provide actionable insights for stakeholders in the EV industry.

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3. Methodology

The analysis followed these steps:

- **Data Collection**: Sales data was obtained from official EV registration records and compiled into a structured dataset.
- **Data Cleaning**: Missing values and inconsistencies were addressed for accurate reporting.
- **Visualization**: Power BI was used to create interactive dashboards showing sales by year, state, and vehicle category.

- **Trend Analysis**: Year-over-year growth rates and category performance were examined.
- **Insights Compilation**: Observations were summarized to support business and policy recommendations.

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4. Dataset Overview

The dataset covers EV sales across **all Indian states and union territories** from **2014 to 2024**. Key data fields include:

- **Year** – Year of sales.
- **State** – State or Union Territory name.
- **Vehicle Category** – Two-wheelers, three-wheelers, and four-wheelers.
- **Sales Quantity** – Number of EV units sold.

Total Sales: **4 million EVs** over the recorded period.

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5. Analysis & Findings

a. Sales Trend Over Time

- **2014–2016**: Slow start, with sales between **0.04M and 0.09M units annually**.
- **2017–2020**: Moderate growth, peaking at **0.17M units in 2019** before a dip in 2020.
- **2021–2023**: Rapid expansion — **1.02M units in 2022** and **1.53M in 2023**, marking the highest-ever sales.
- **2024 (partial)**: 0.14M units sold so far.

b. Top Performing States

1. **Uttar Pradesh** — 0.73M units
2. **Maharashtra** — 0.40M units
3. **Karnataka** — 0.32M units
4. **Rajasthan** — 0.23M units
5. **Gujarat** — 0.18M units

These states have benefited from strong government support, better charging infrastructure, and higher urban EV adoption rates.

c. Vehicle Category Distribution

- **2-Wheelers** — 1.81M units (45% of total sales)
- **3-Wheelers** — 1.62M units (40% of total sales)
- **4-Wheelers** — 0.15M units (4% of total sales)

2-wheelers dominate the market due to affordability and suitability for urban commuting, while 3-wheelers play a significant role in commercial and shared mobility.

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6. Conclusion

The analysis reveals that India's EV market has experienced a dramatic growth surge in recent years, especially after 2021. The adoption pattern shows:

- Dominance of **2- and 3-wheelers** in the EV market.
- Certain states like Uttar Pradesh and Maharashtra leading in sales.
- Strong growth momentum that is likely to continue with government support and infrastructure development.

The insights from this project can assist in shaping EV policy, expanding infrastructure, and encouraging investments in emerging EV technologies.